OSMatrix

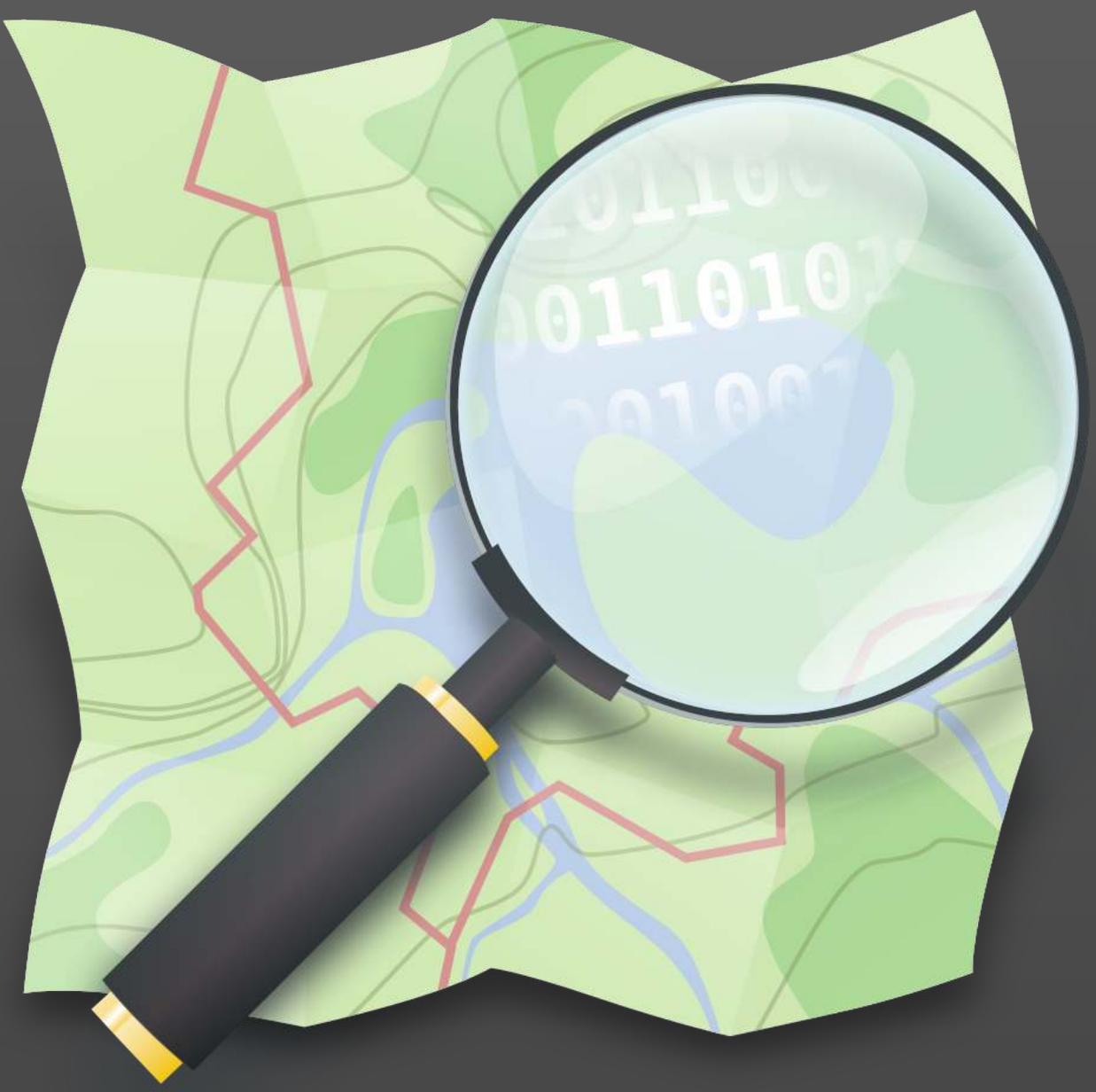
Grid-based Analysis and
Visualization of OpenStreetMap

Oliver Roick
Julian Hagenauer
Alexander Zipf

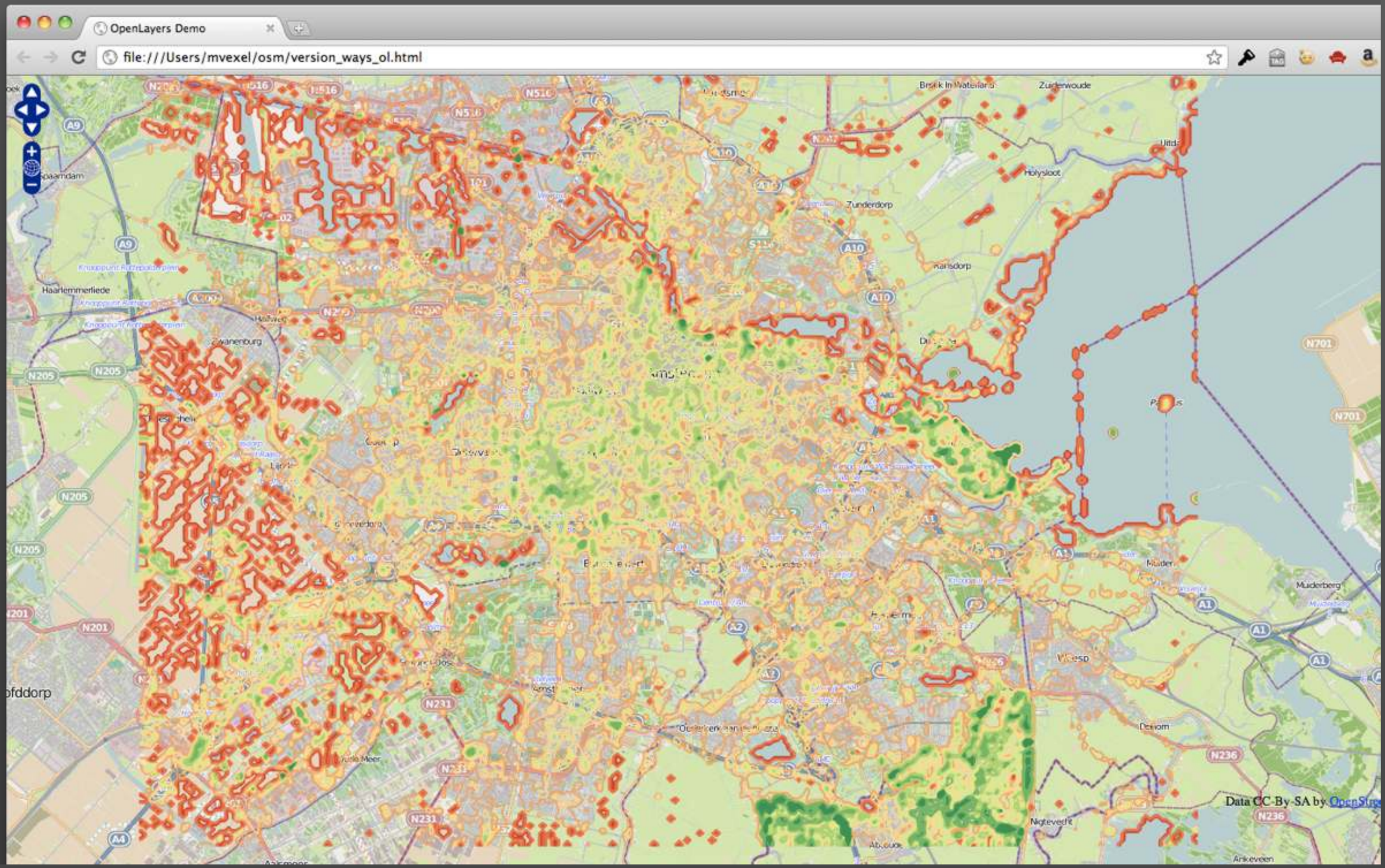
What is OSMatrix?


THE MATRIX

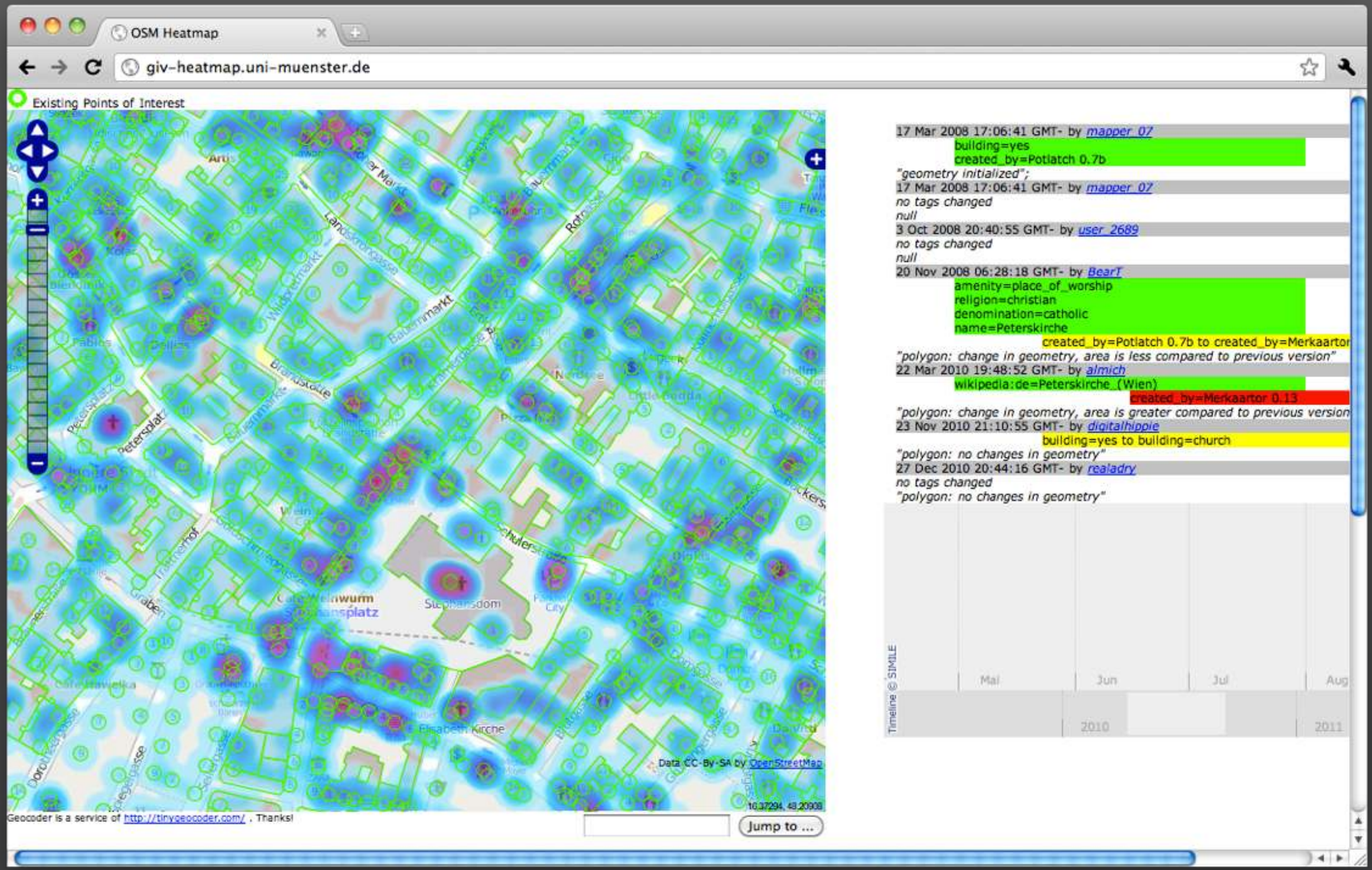
What is OSMatrix?



Martijn van Exel | wp.me/pcUHq-5f



Johannes Trame & Carsten Keßler | giv-heatmap.uni-muenster.de



Min/Max/Avg:

- Version number
- Number of Contributions per User
- Number of attributes

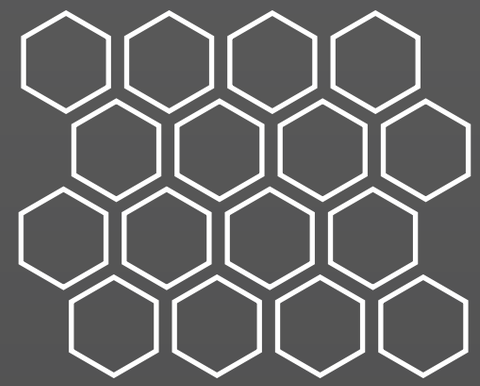
Sum:

- Attributes
- Features
- Contributing users

Area:

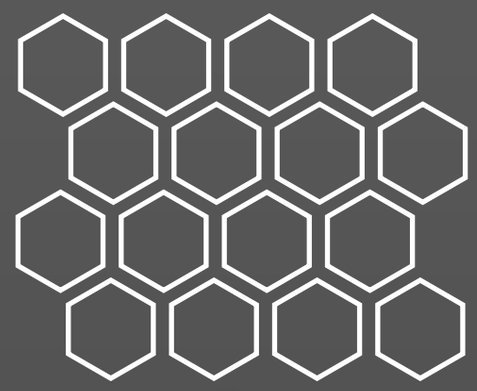
- Buildings
- Landuse

...



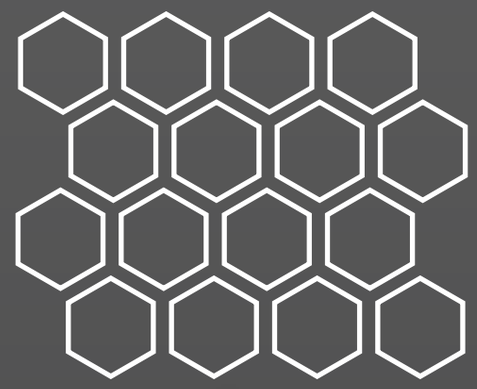



OSM





Geotools

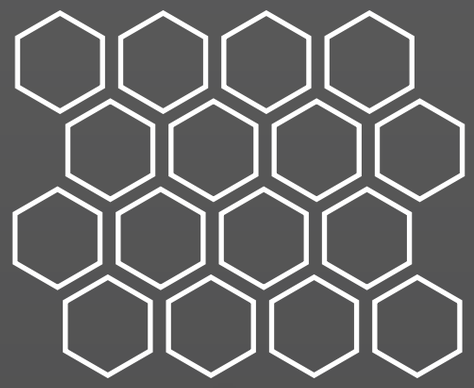






PostGIS

Geotools





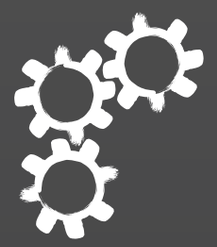
Server



Geoserver



PostGIS



Geotools

Client

OpenLayers

Server

Geoserver

PostGIS

OSM

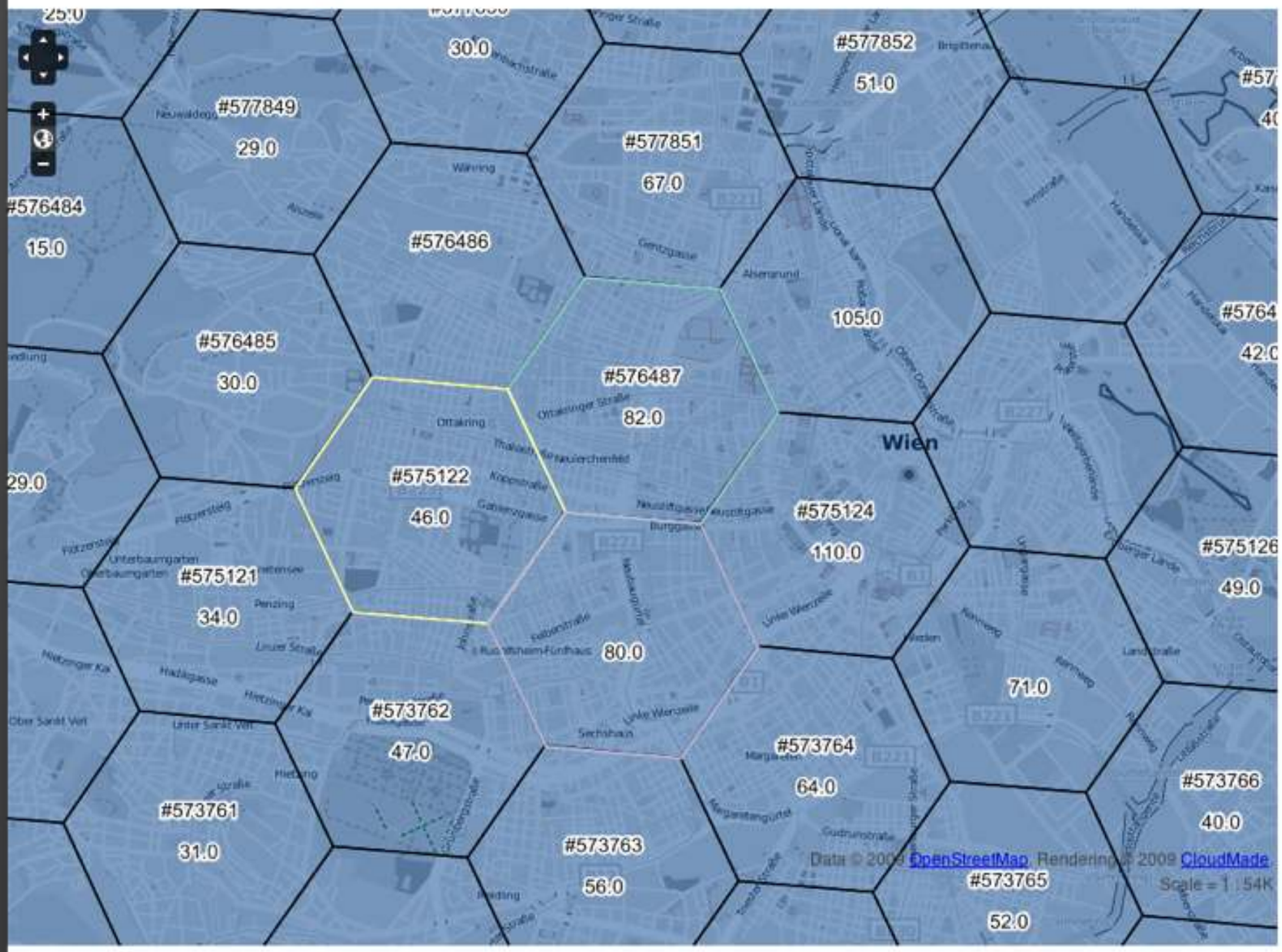
Geotools

OSMatrix

koenigstuhl.geog.uni-heidelberg.de/osmatrix/?layer=osmatrixeu:numuser&zoom=13&center=16.33987422357585,48.20829560126006#

OSMatrix

Go to my location | Find place | Permalink | Additional information



Switch background layer | Choose attribute

Number of contributing users

Overall number of users that modified an object within the given cell.

- 0]
-]0 - 1]
-]1 - 2]
-]2 - 3]
-]3 - 4]
-]4 - 6]
-]6 - 10]
-]10
- Other

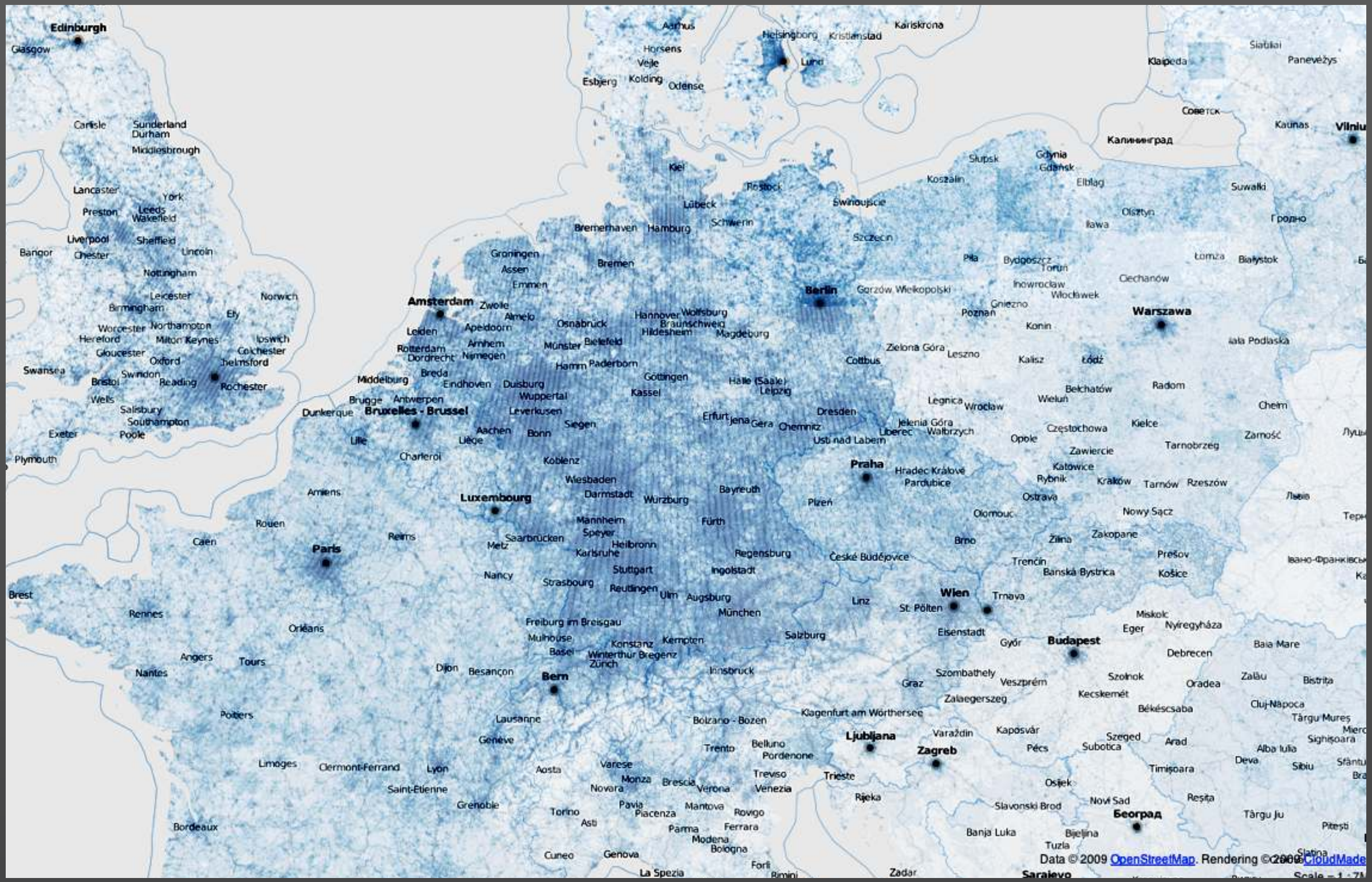
Feature Info

Cell ID	Value
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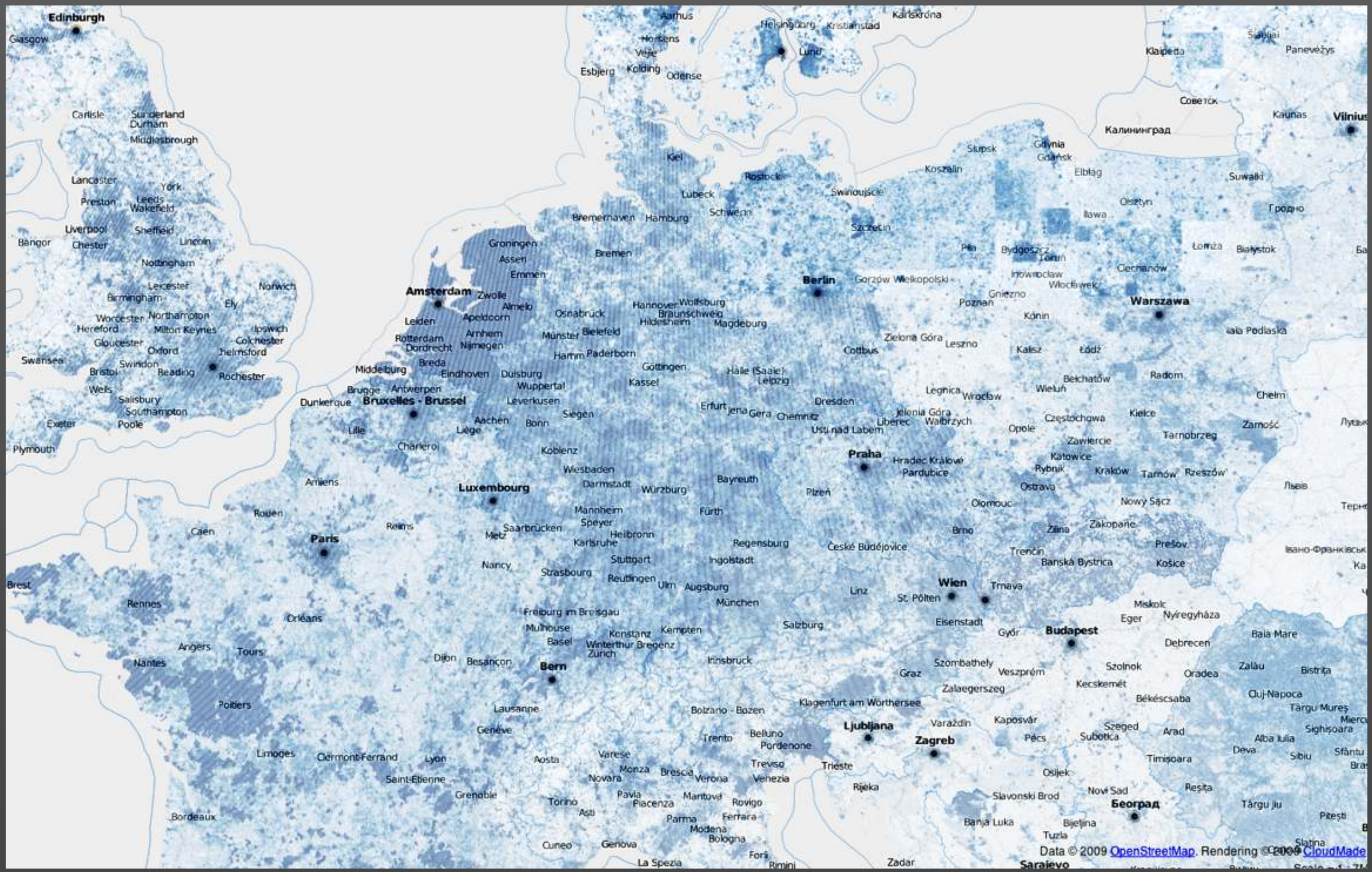
GIScience Heidelberg • Maps and data: © OpenStreetMap and contributors, CC-BY-SA

Results

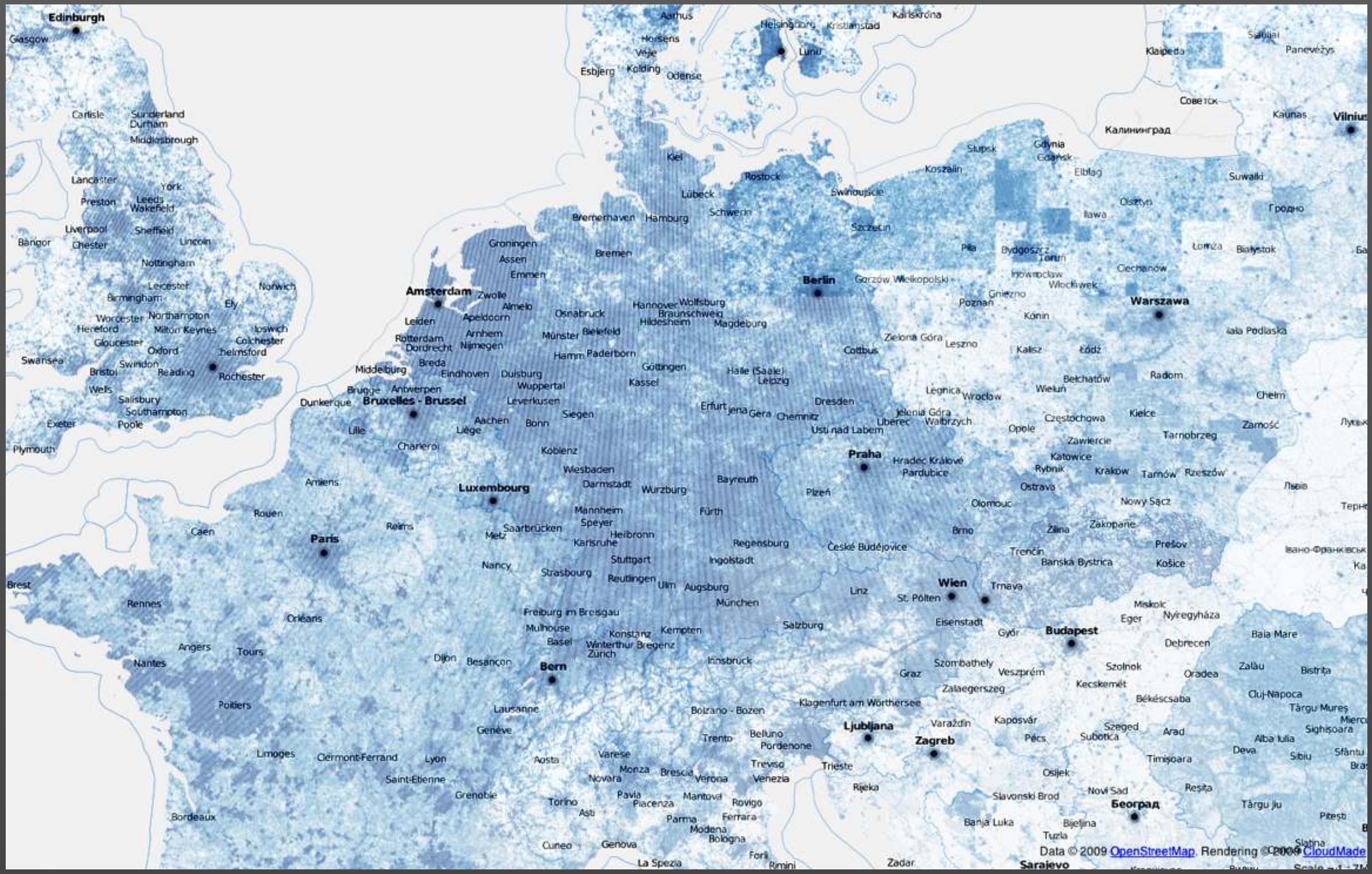
Number of contributing users



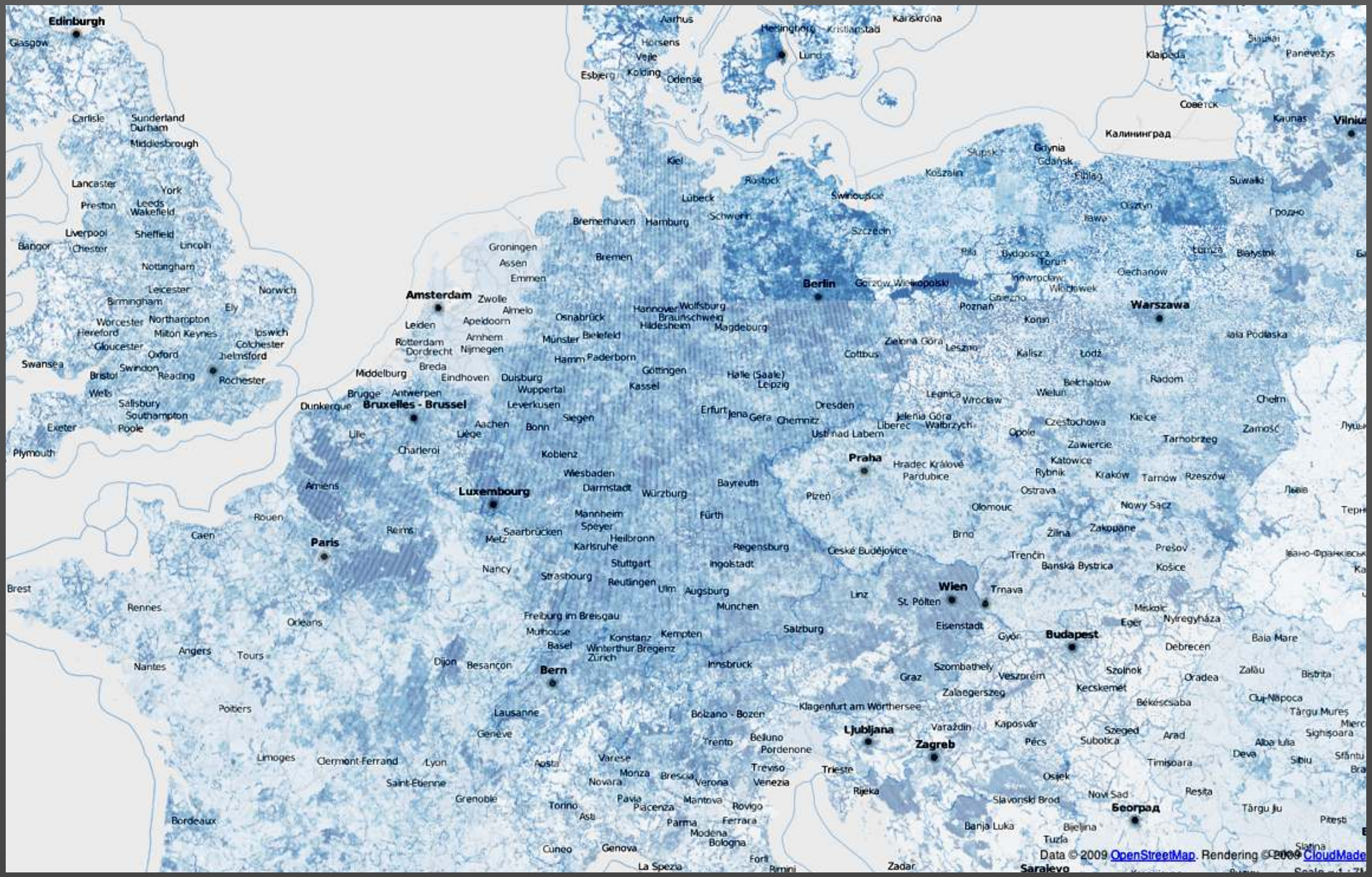
Average number of contributions per user



Total number of features

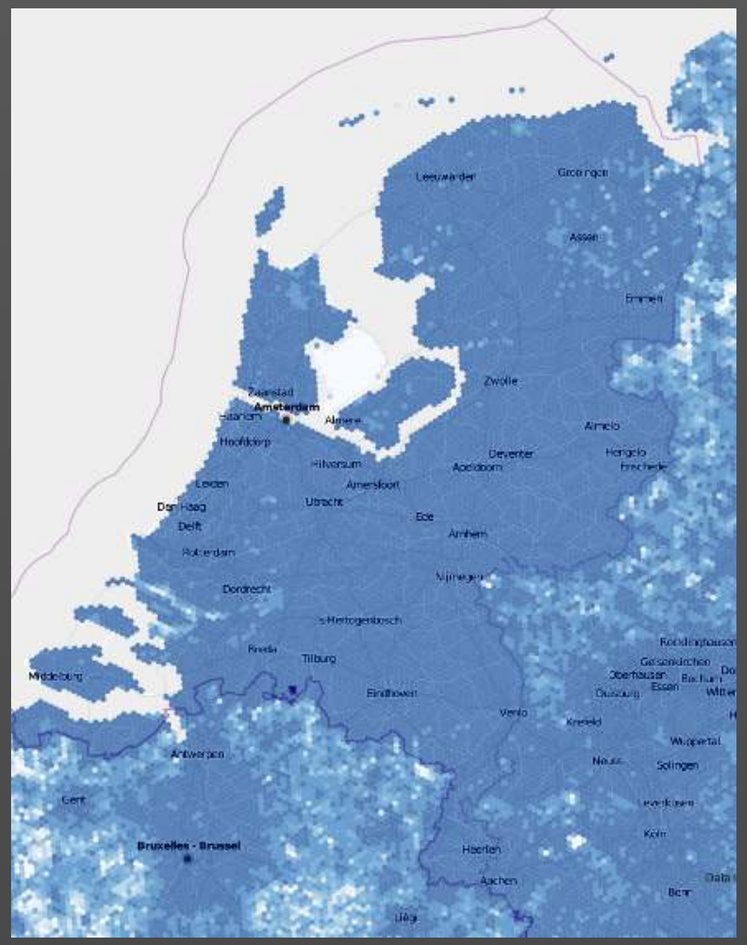


Average version number

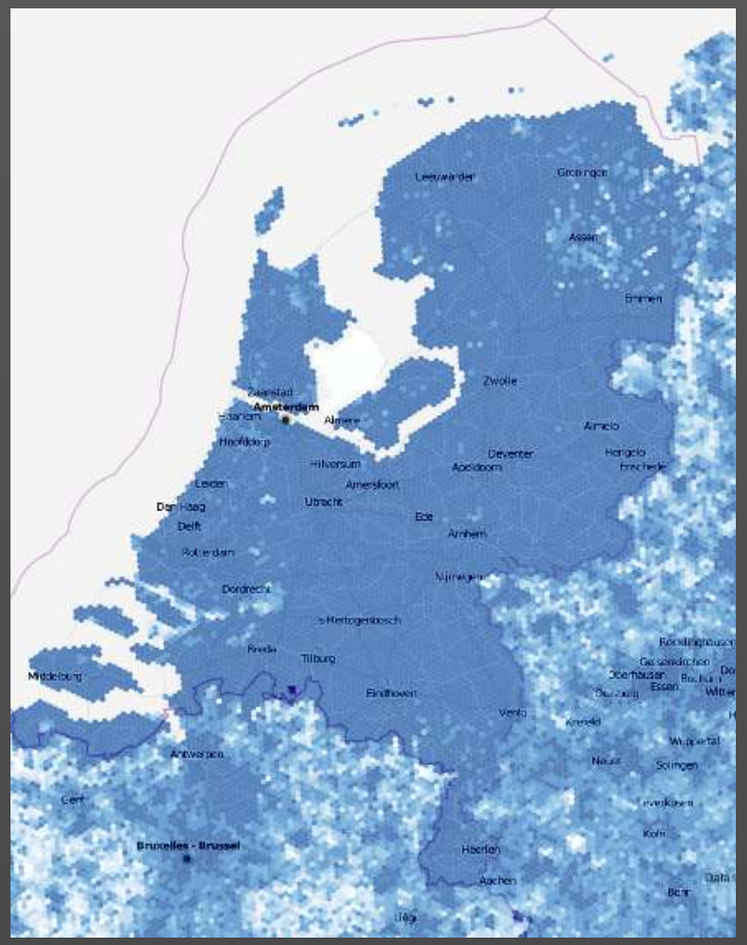

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Lincoln
Nottingham
Leicester
Norwich
Birmingham
Ely
Worcester
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Colchester
Oxford
Swansea
Swindon
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Wells
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Southampton
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Legnica
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Hradec Králové
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The Netherlands

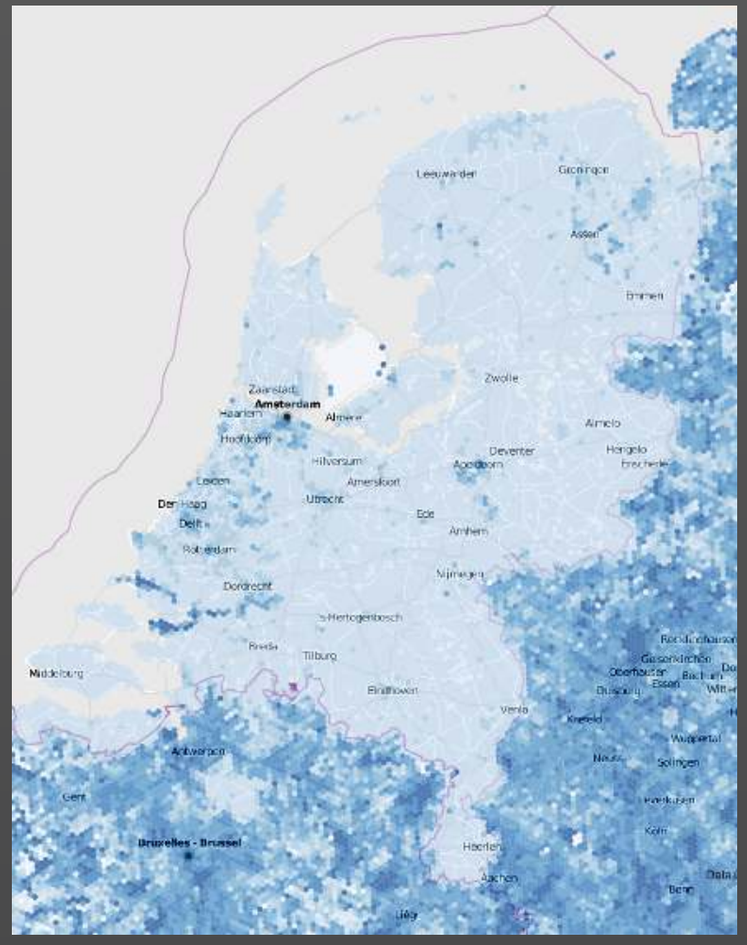
Number of features



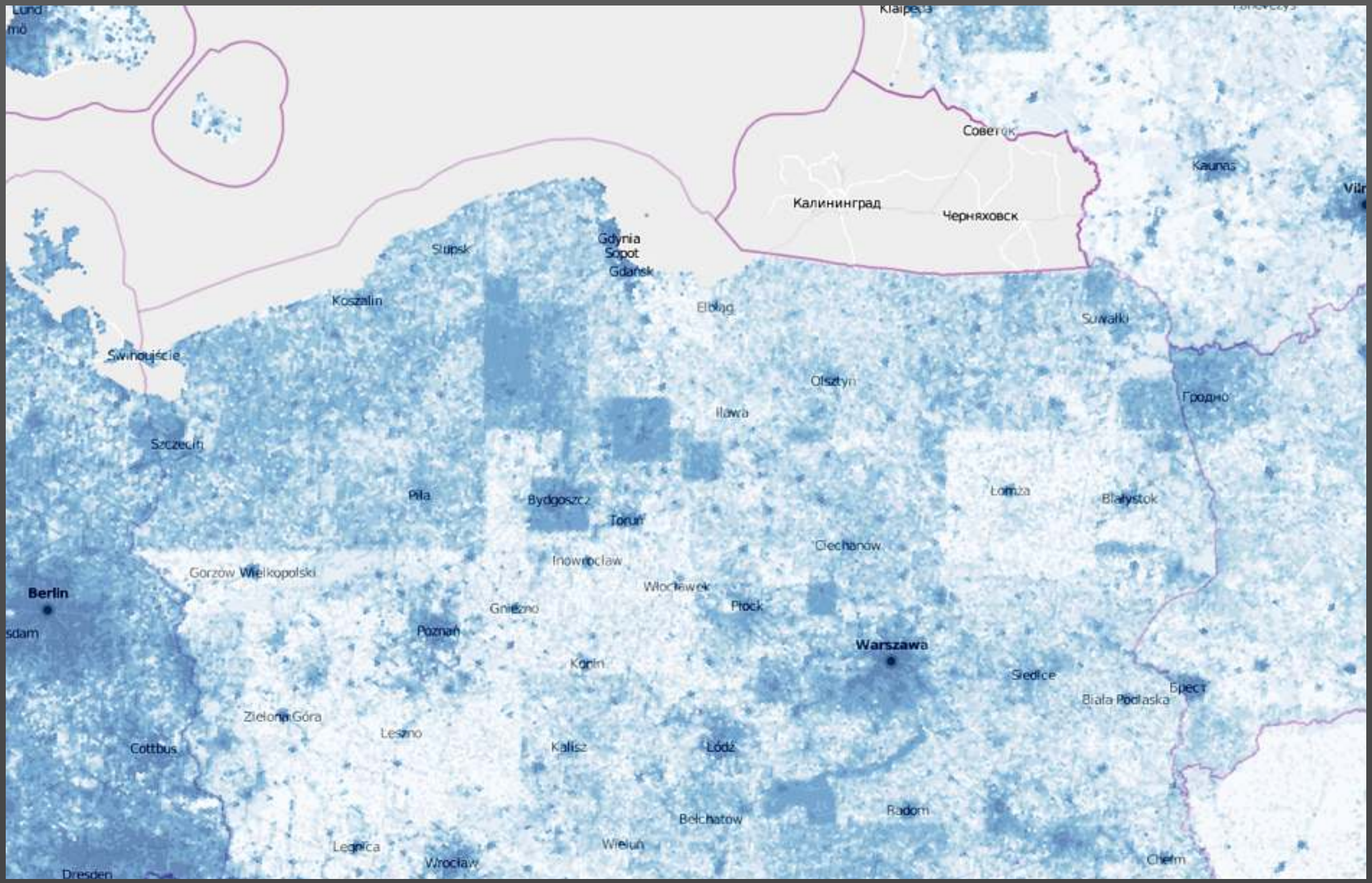
Average number of contributions per user



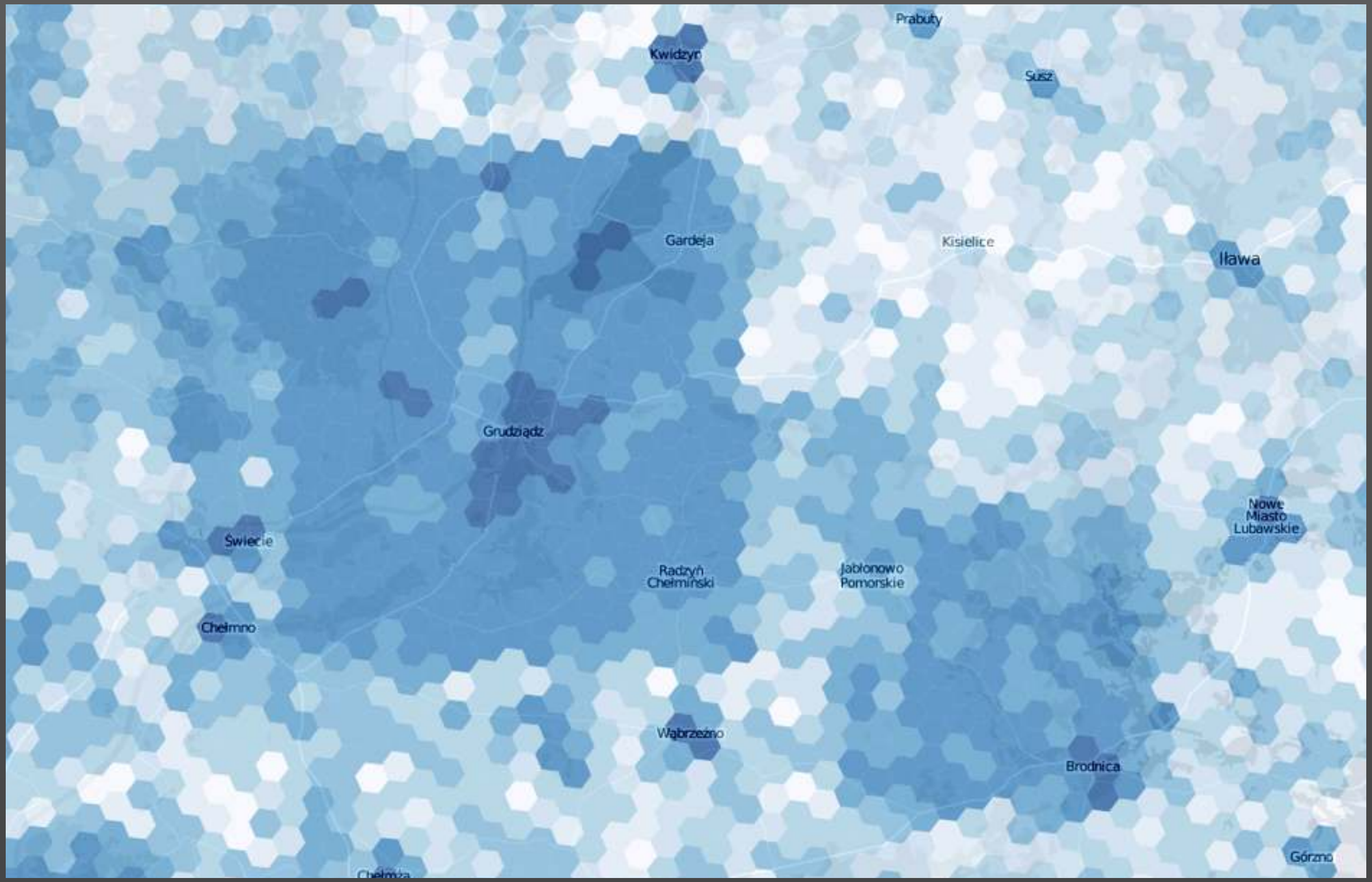
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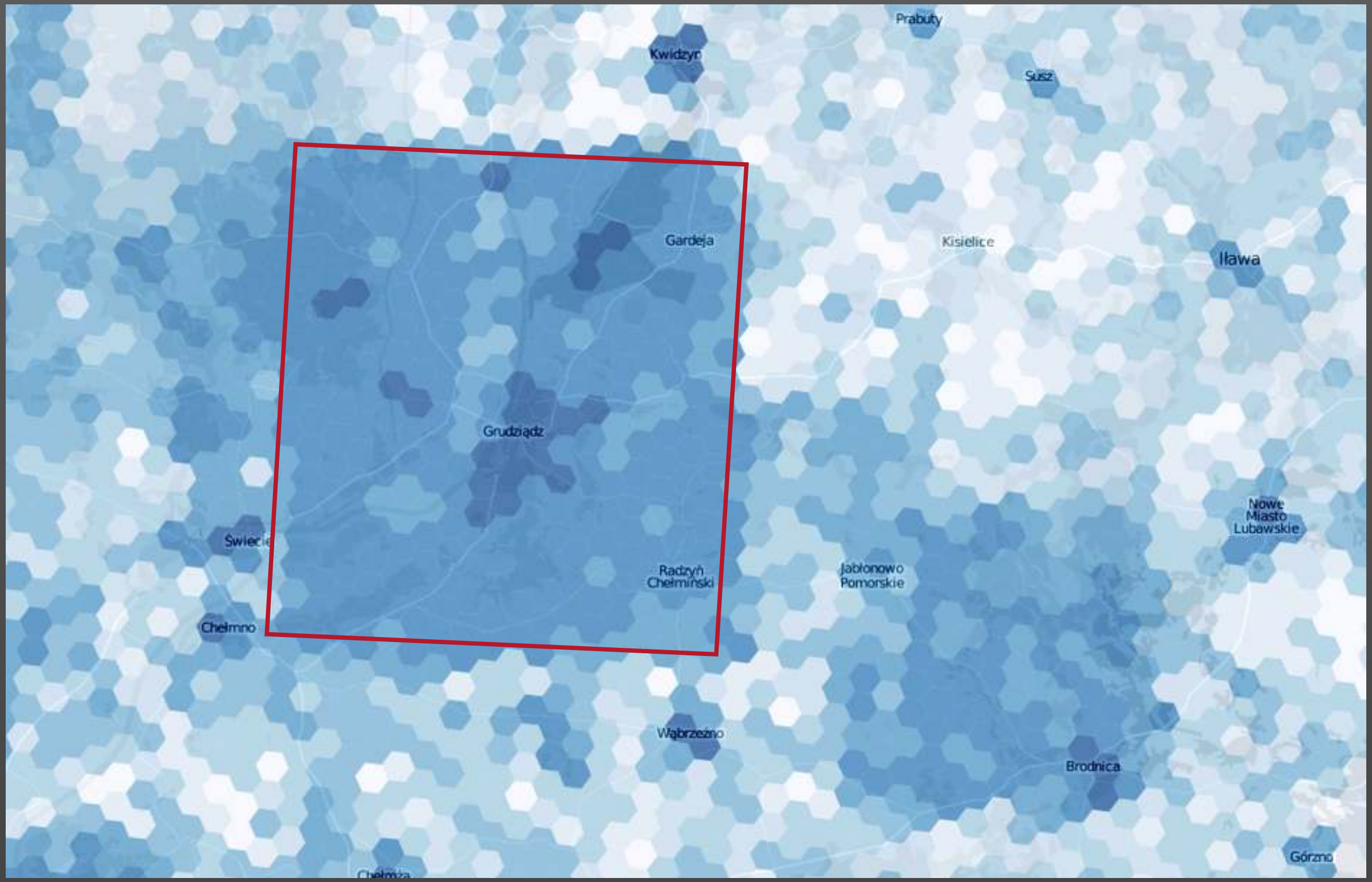
Poland | Total number of features



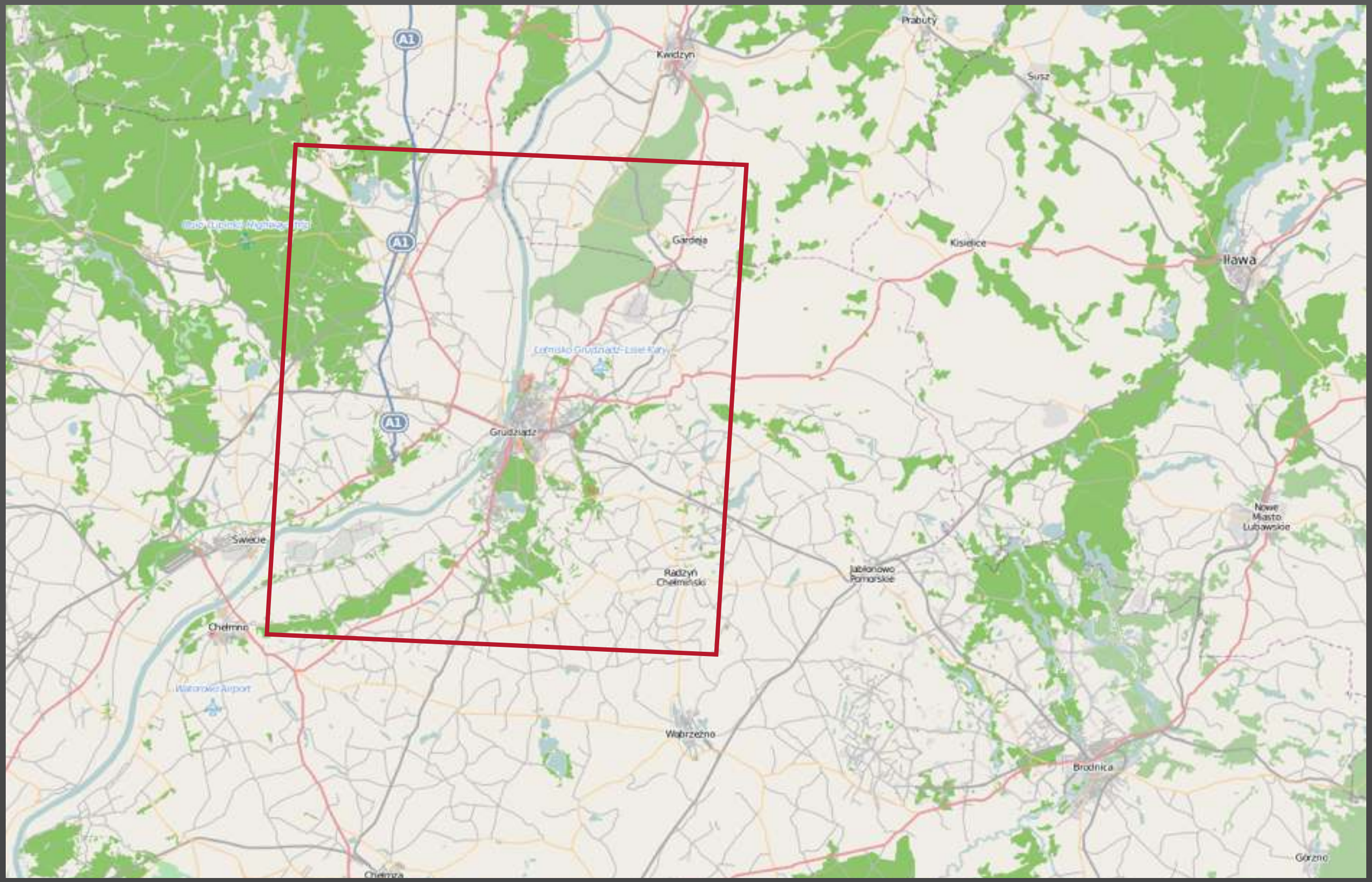
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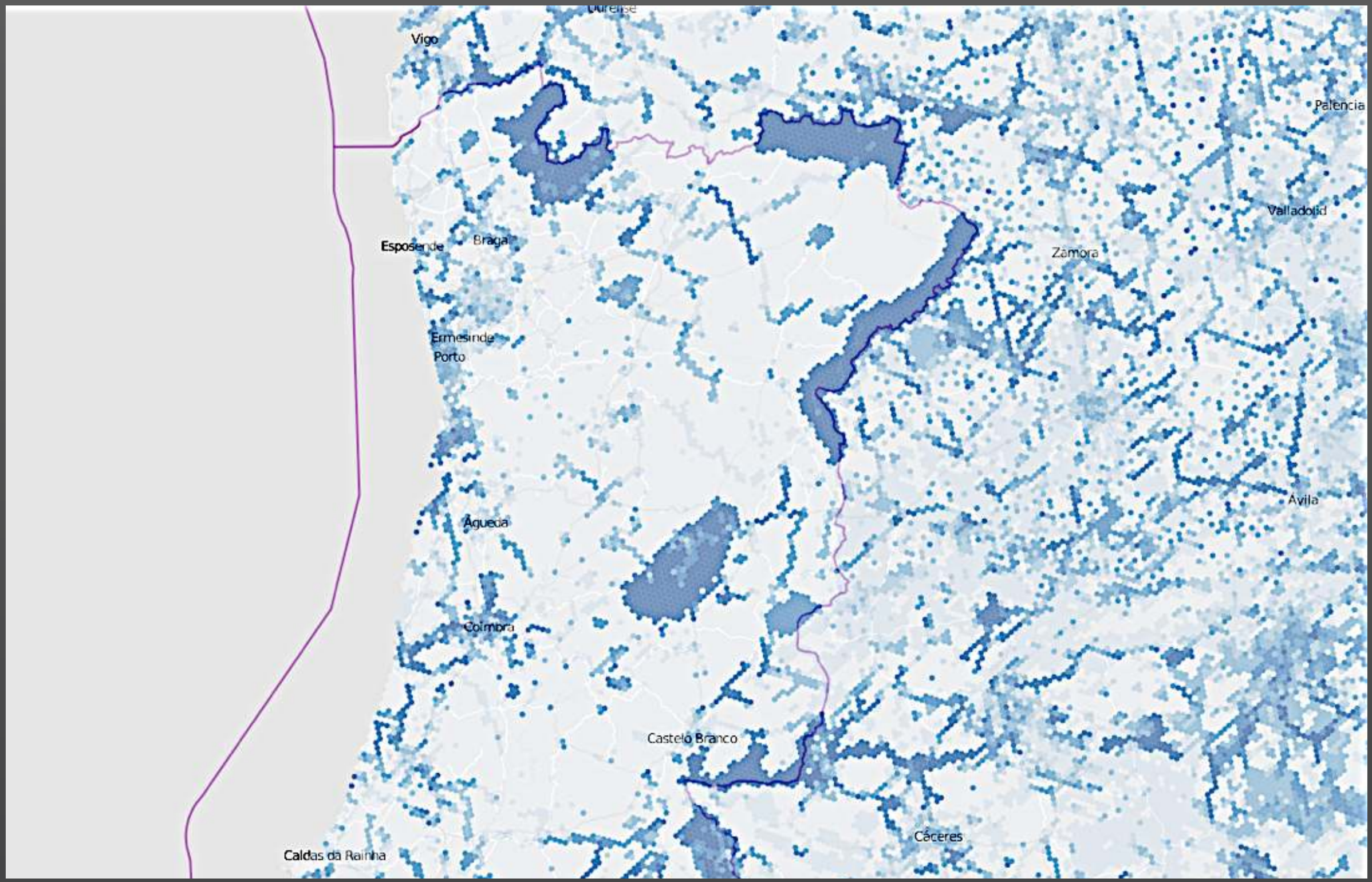
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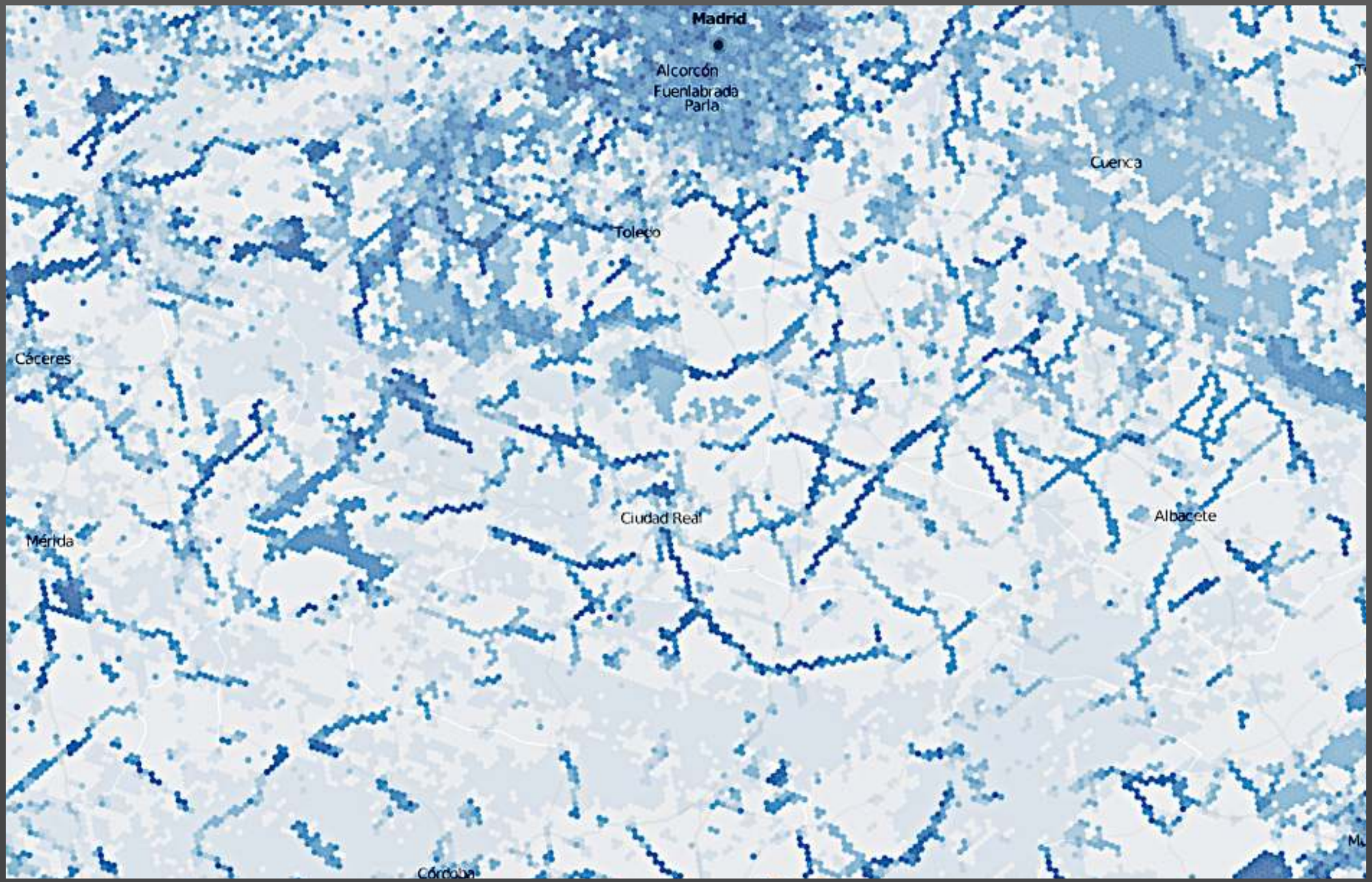
Poland | Total number of features



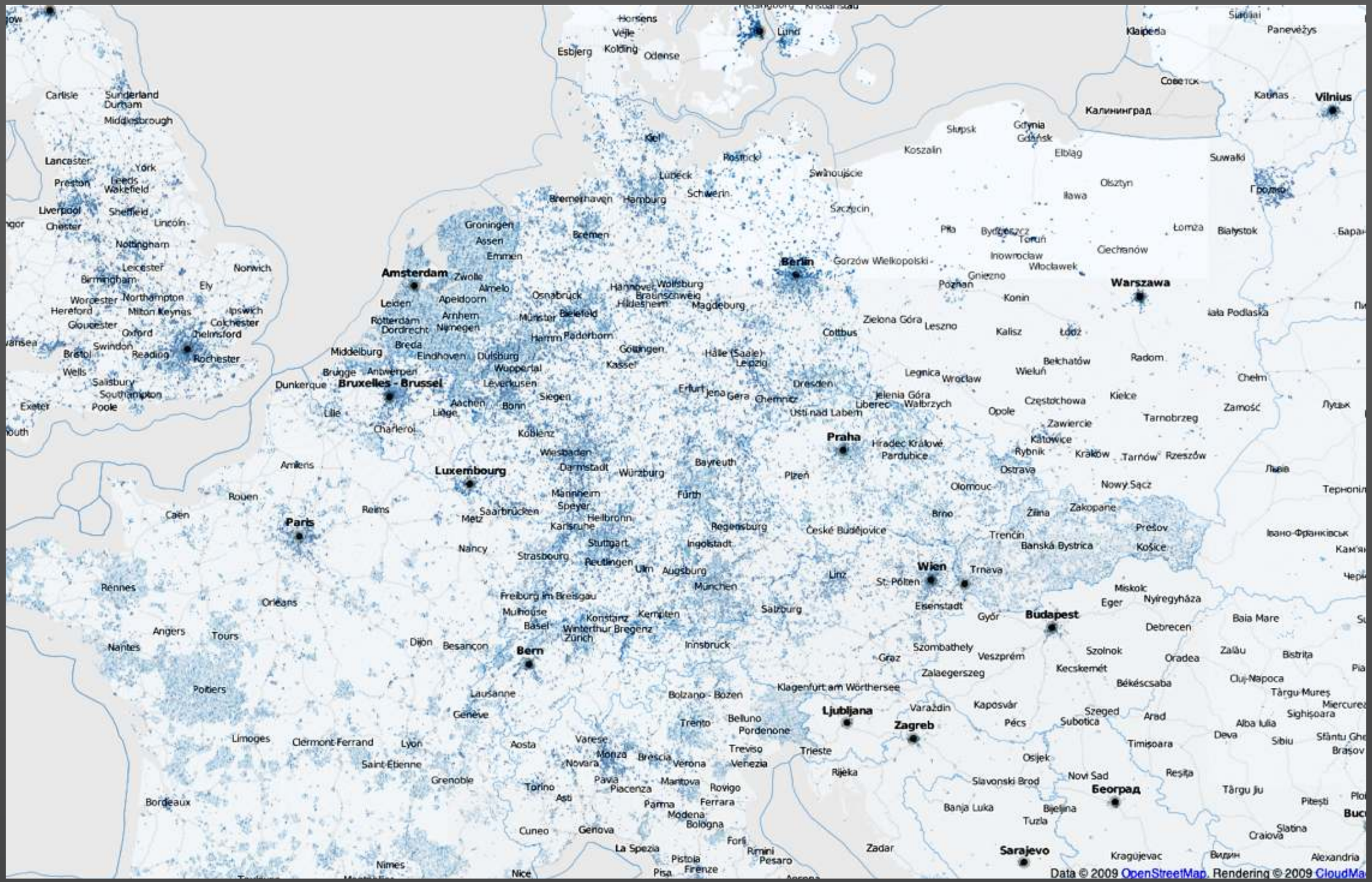
Portugal | Average version number



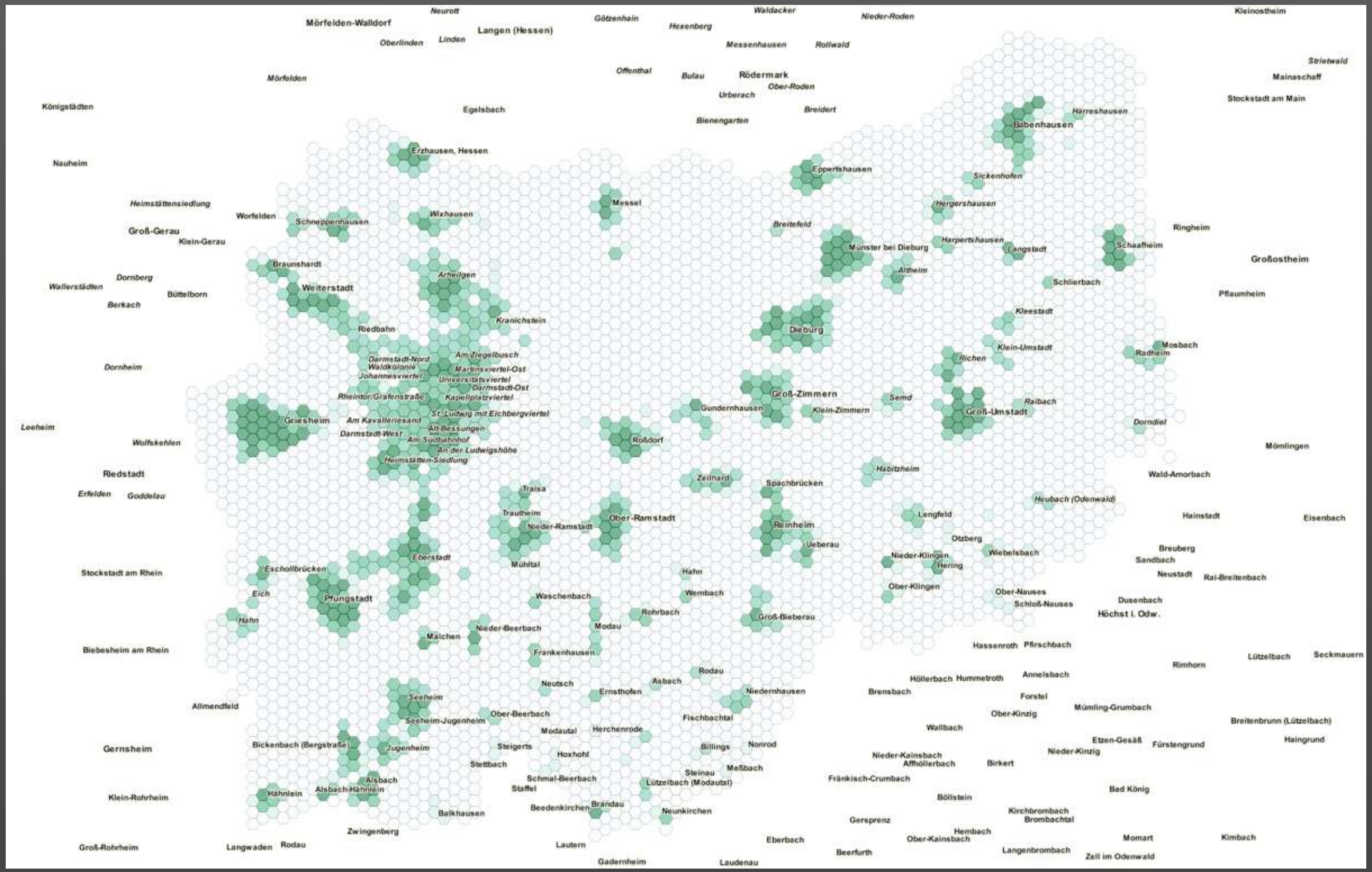
Spain | Average version number



Area covered by buildings

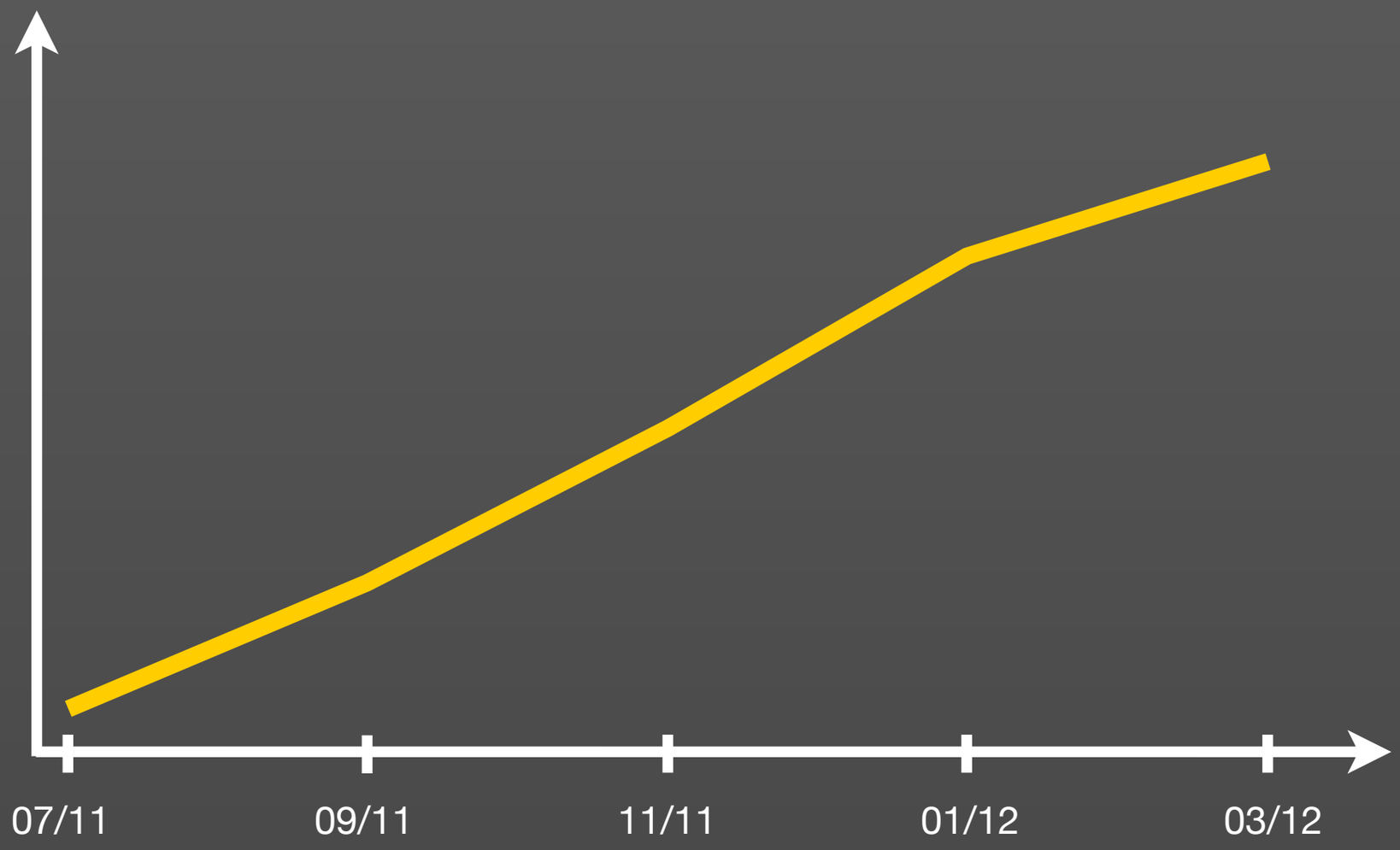


Julian Hagenauer & Marco Helbich



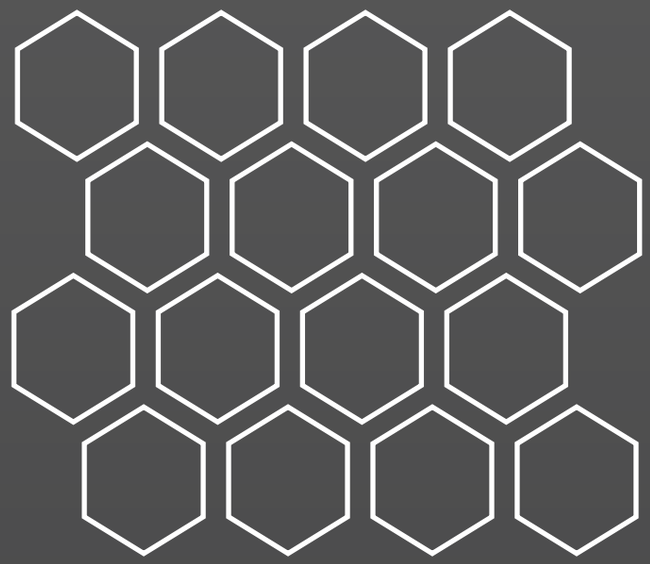
What‘s next?

Temporal information

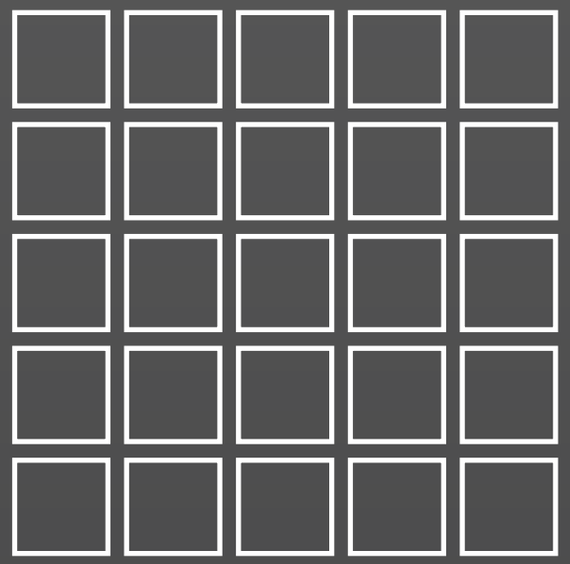


Improve performance

Influence of cell size and shape



?



Thank you.
Questions?

http://osmatrix.uni-hd.de

Oliver Roick

Ruprecht-Karls-Universität Heidelberg
Institute of Geography
Chair of GIScience

roick@uni-heidelberg.de

twitter.com/oliverroick
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